Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/27/2021
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2016	**End Date:**	09/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2022 per NSSC information (Ed., 9/22/21)		
Key Personnel Changes/Previous PI:	June 2019 report: No change to report.		
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX16AT09G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: The NASA Task Book reflects the NSSC project title, "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism." The project is also known as "Electrically Driven Liquid Film Flow Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism" (Ed., 7/18/23). ---------- NOTE: Continuation with same Principal Investigator (PI) Jamal Yagoobi, Ph.D. of "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism," grant NNX12AR32G. "Electrically Driven Liquid Film Flow Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction		

Task Description:	Mechanism" is an International Space Station (ISS) Experiment to fundamentally investigate the effects of gravity on: 1.) electrically generated two-phase flow (liquid/vapor) and 2.) electrically driven liquid film flow boiling. The proposed work is a collaborative effort between the investigators to develop and validate the flight experiment hardware. We propose this effort based upon the successes of our current project: NASA Grant/Cooperative Agreement: NNX12AR32G Electrically Driven Liquid Film Flow Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism. Although the original effort extends through February 2017, we have completed tasks and achieved the technical and programmatic milestones of this project. That is, we have passed all the requirements of the Science Concept Review and received the authorization to advance to the next stage of the experiment development. Therefore, the PI and Co-PI submit this proposal to develop the flight experiment hardware with an effective start date of October 2016. This proposal fundamentally investigates the complex liquid-vapor phase change phenomenon in presence of electrical field and absence of gravity. Thus, this study will provide valuable scientific understanding about competing physical phenomena of flow inertia and bubble dynamics in electrically driven liquid film flow boiling in the presence of electric bubble extraction force (i.e., dielectrophoretic force) in microgravity. This is a proposal to support the development of the experiment flight hardware for the International Space Station, conduct the experiment on–board the International Space Station, support the operations of the experiment and evaluate the experimental data. The proposed period of performance is October 2016 to September 2021. The robust collaboration between Worcester Polytechnic Institute (WPI) and Goddard Space Flight Center (GSFC) can provide both state of the art theoretical investigation and deliver flight qualified experimental hardware; thus, this proposal presents both investigative and hardware development tasks. The PI will maintain overall technical and scientific direction of the experiment development; however, the Co-I will support the flight experimental hardware development in collaboration with Glenn Research Center. Note: This project successfully passed all requirements of the Science Concept Review and has received authorization to advance to the next stage of development. This is to elevate it to the International Space Station flight status as of October 1, 2016, with this continuation grant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This multi-physics proposal will provide basic scientific understanding about competing physical phenomena of flow inertia and bubble dynamics in electrically driven liquid film flow boiling in the presence of electric bubble extraction force (i.e., dielectrophoretic force) in microgravity. The electrically driven liquid film flow boiling data aboard ISS will determine the critical heat flux, onset of boiling, absence of hysteresis, and bubble characteristics in the absence of gravity. There are a number of potential terrestrial and space applications for this electrohydrodynamic (EHD) driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices. To achieve these applications, the basic scientific understanding about competing physical phenomena of flow inertia and bubble dynamics in electrically driven liquid film flow boiling in the presence of electric bubble extraction force (i.e., dielectrophoretic force) in microgravity is needed.
Task Progress:	Detailed experimental work was carried out in the Multi Scale Heat Transfer Laboratory at Worcester Polytechnic Institute to understand the following, a-The effect of EHD conduction pumping on liquid film flow boiling, b-The effect of dielectrophoretic (DEP) force on liquid film flow boiling, c-The combined effect of EHD conduction pumping and DEP force on liquid film flow boiling. The results for DEP case show a clear enhancement in heat transfer. This is characterized by reaching higher heat fluxes for the same temperature difference or lower temperature differences for the same heat flux value. The results further improve with the combined effect of DEP plus EHD. This is true with the critical heat flux values as well. The dielectric constant (i.e., normalized electrical permittivity of the fluid with respect to vacuum electrical permittivity) and electric conductivity of the working fluid HFE 7100 were measured as a function of temperature in the Multi Scale Heat Transfer Lab using the unique facility. The dielectric constant decreases gradually with temperature which is in agreement with those of other dielectric fluids reported in the literature. The electric conductivity of HFE 7100 increases initially with temperature, reaching its peak value at about 15 C. It then drops with the increase in temperature. This trend is different for this fluid compared to the most other dielectric fluids. However, our results are very similar to those reported by the University of Poitiers, France. The thermal analysis of electrically driven liquid film flow boiling experiment chamber in the absence of gravity was continued in this period as well. This effort was necessary to address various safety concerns that were raised throughout the year. This task may continue in the upcoming year as well. The analyses employ COMSOL Finite Element Analysis (FEA) software version 5.4. A unique test unit was fabricated to study a potential incompatibility concern with the coating that will be applied to the inner surface of the glass to be installed on the upper surface of the experiment chamber. The results, thus far, show no incompatibility issue with the working fluid HFE 7100. Thermoelectric Cooler (TEC) performance to be used for cooling of the experiment chamber onboard ISS were evaluated in the Multi Scale Heat Transfer Lab. The results indicated that TEC is capable of providing reliable cooling capacity for the ISS chamber.
Bibliography Type:	Description: (Last Updated: 01/30/2026)

Abstracts for Journals and Proceedings	Castaneda AJ, O'Connor NJ, Yagoobi J. "Investigation of Gravity Effects on Electrically Driven Liquid Film Flow. " 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Articles in Peer-reviewed Journals	Talmor M, Yagoobi J. "Numerical study of micro-scale EHD conduction pumping: The effect of pump orientation and flow inertia on heterocharge layer morphology and flow distribution control." Journal of Electrostatics. 2021 May;111:103548. https://doi.org/10.1016/j.elstat.2020.103548 , May-2021
Papers from Meeting Proceedings	Castaneda AJ, O'Connor NJ, Yagoobi J. "Investigation of Gravity Effects on Electrically Driven Liquid Film Flow Boiling: A Micro-Gravity Flight Campaign in Preparation of ISS Experiment." IMECE® 2020 International Mechanical Engineering Congress & Exposition, Virtual, November 16 – 19, 2020. Proceedings. IMECE® 2020 International Mechanical Engineering Congress & Exposition, Virtual, November 16 – 19, 2020. Paper No: IMECE2020-24133, V011T11A008. https://doi.org/10.1115/IMECE2020-24133 , Nov-2020
Papers from Meeting Proceedings	Nishikawara M, Yagoobi J. "Experimental Study of Electrohydrodynamic Conduction Pumping Embedded in Micro-Scale Evaporator." 2020 IEEE Industry Applications Society Annual Meeting, Virtual, Detroit, MI, October 10-16, 2020. 2020 IEEE Industry Applications Society Annual Meeting. Proceedings. https://doi.org/10.1109/IAS44978.2020.9334813 , Oct-2020
Patents	63107042. Provisional patent issued October 2020. Oct-2020 Yagoobi J. "Gravity Independent Liquid Cooling for Electronics."